

A simple visual heuristic allows maze navigators to choose shorter routes under combinatorial uncertainty

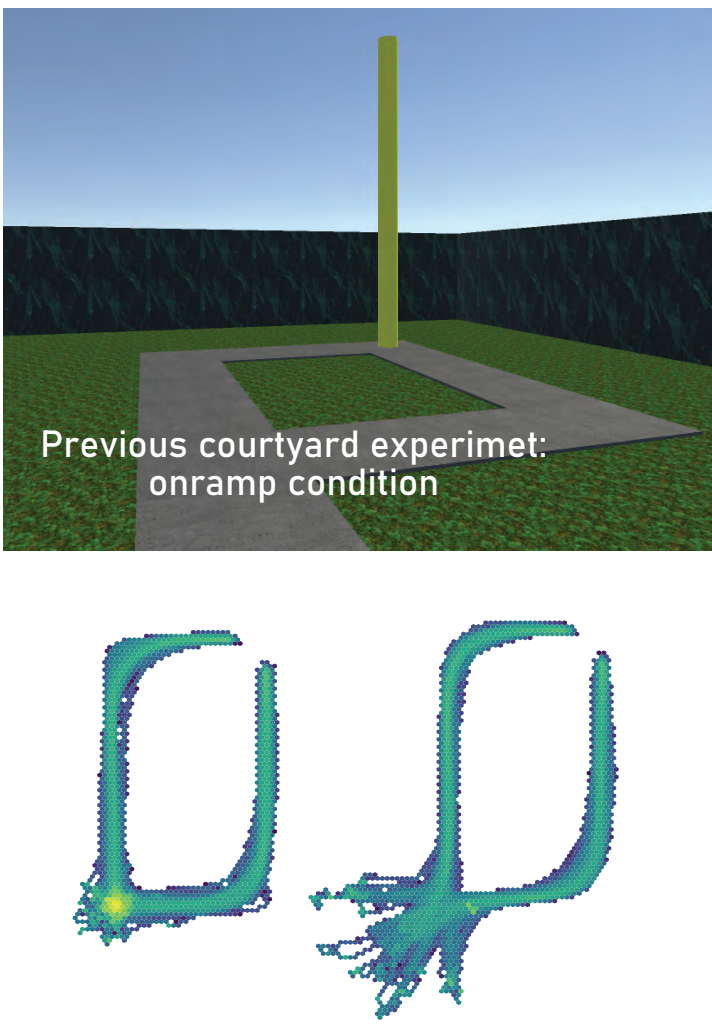
Cassandra Engstrom₁ & William H. Warren₁

BROWN

1 Department of Cognitive and Psychological Sciences, Brown University, Providence, RI, USA, 02912

Background

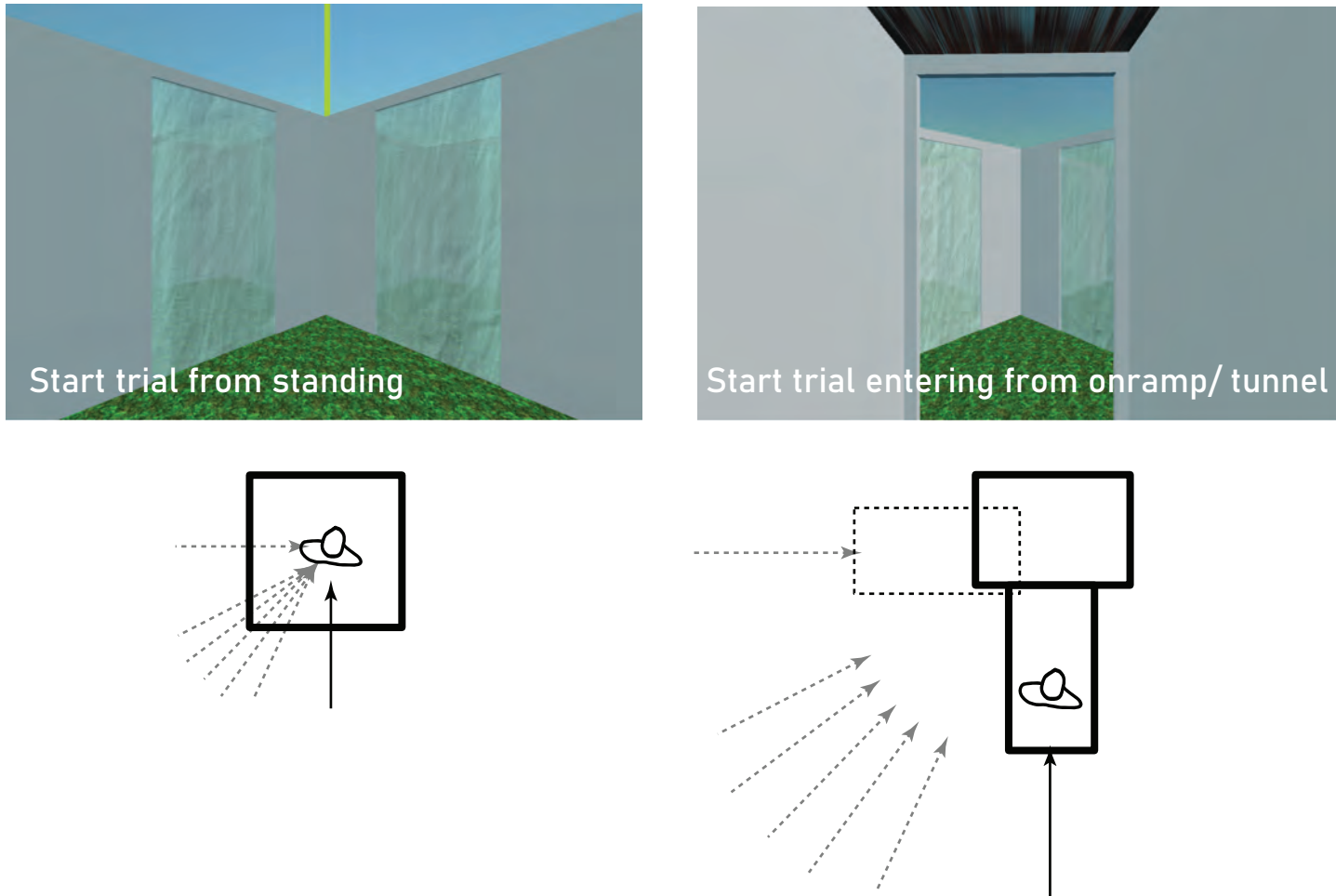
Previous work in our lab has found that humans navigating through unfamiliar mazes avoid deviating from the beeline to their goal destination and their current direction of travel (heading), strategies that can come in conflict [1,2]. While these two factors trade off under uncertainty, neither are influential when routes are visible and equal (see right). To test whether this is because of route occlusion or equality, we designed a grid maze with potentially equal routes and looked for navigational biases.



We found no evidence of a route-based bias for participants walking around a courtyard from varying approach angles

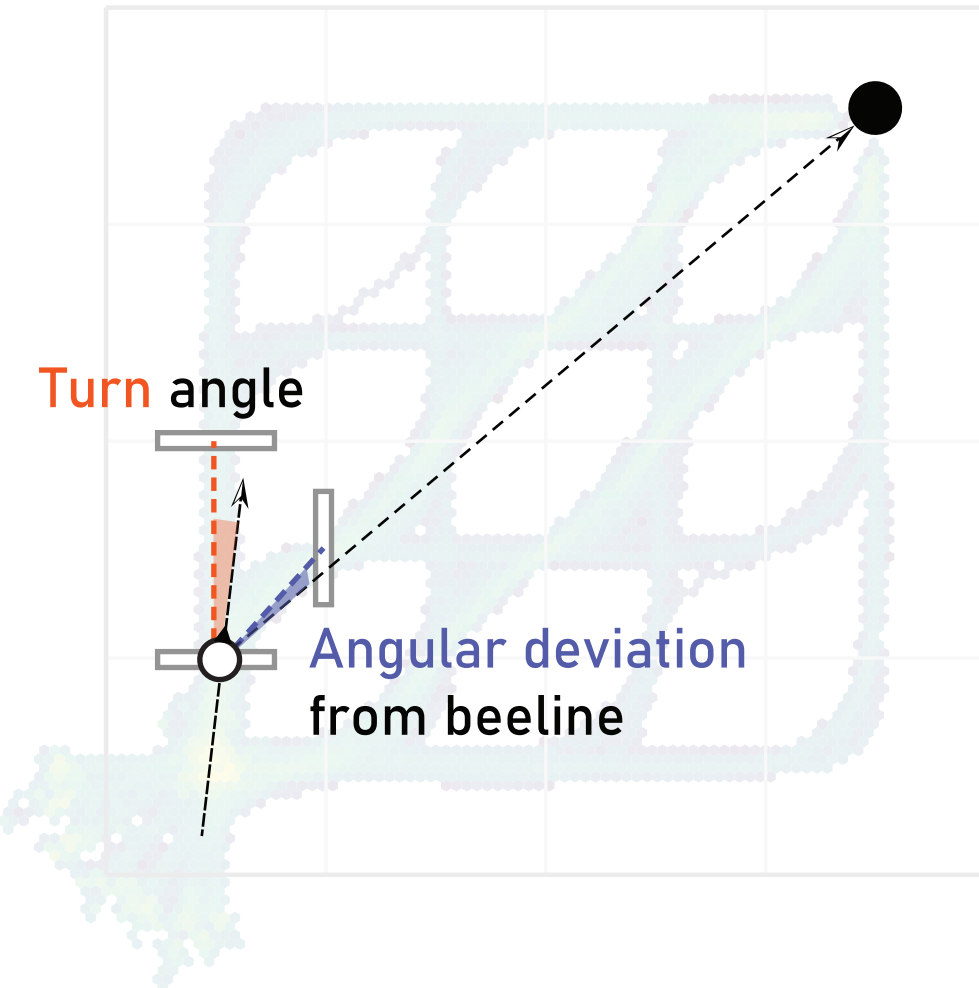
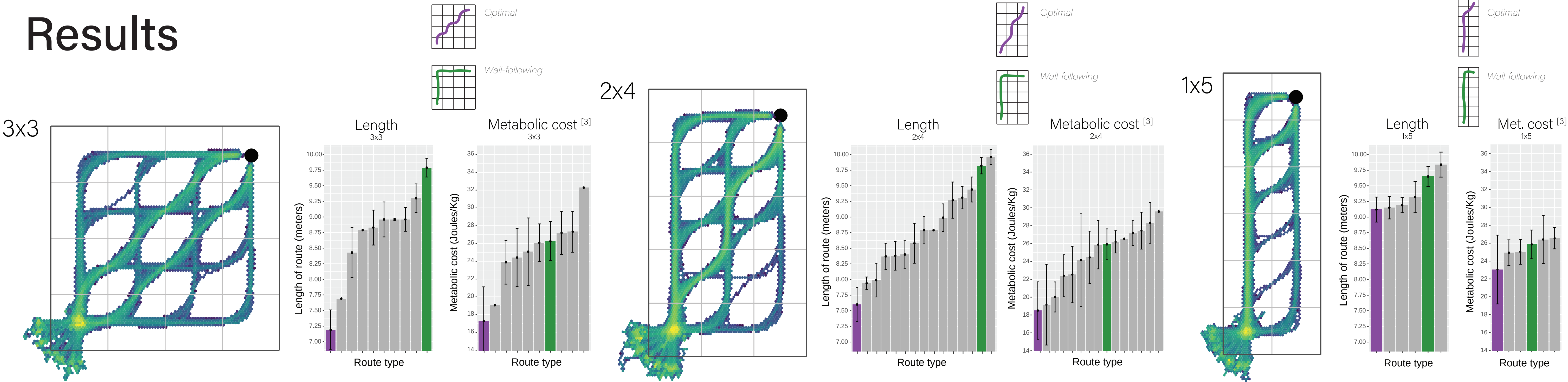
Methods

- 16 participants walked through a 14m x 20m virtual grid-style maze wearing a Quest Pro HMD
- The task was to efficiently walk to a visible beacon
- Mazes had 3 different aspect ratios (ARs) : 1x5, 2x4 and 3x3. Each cell of the grid interior was populated with doors on all sides.
- The experiment had 70 trials, involving all ARs and all entrance conditions, each L/R mirrored.
- Before the maze appeared, participants faced a pole that specified their orientation at trial start.

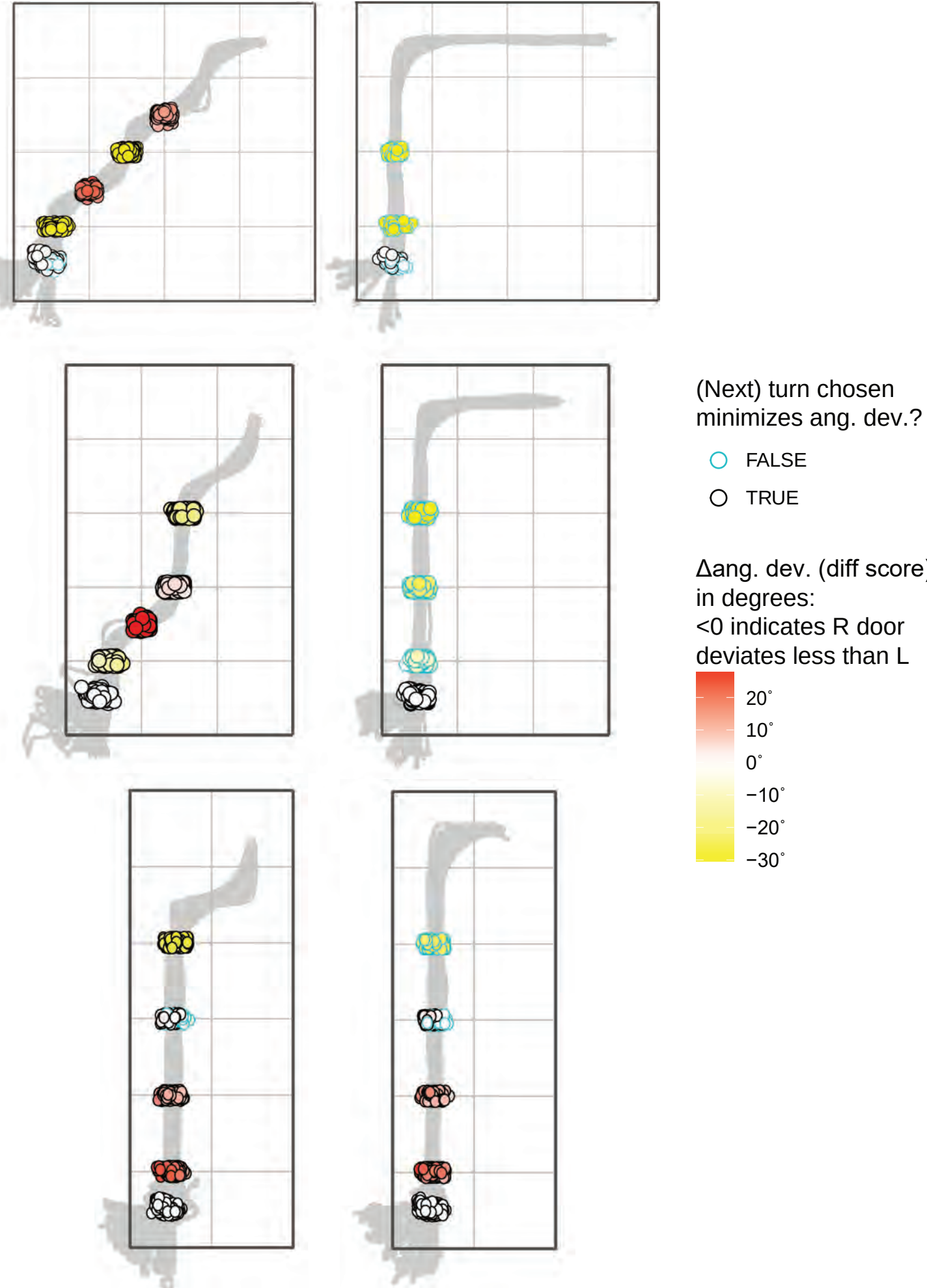


- Participants began each trial facing 1 of 7 different directions. They either started in the center of the first corner cell (left), or they entered the maze from a ~2m tunnel (right).

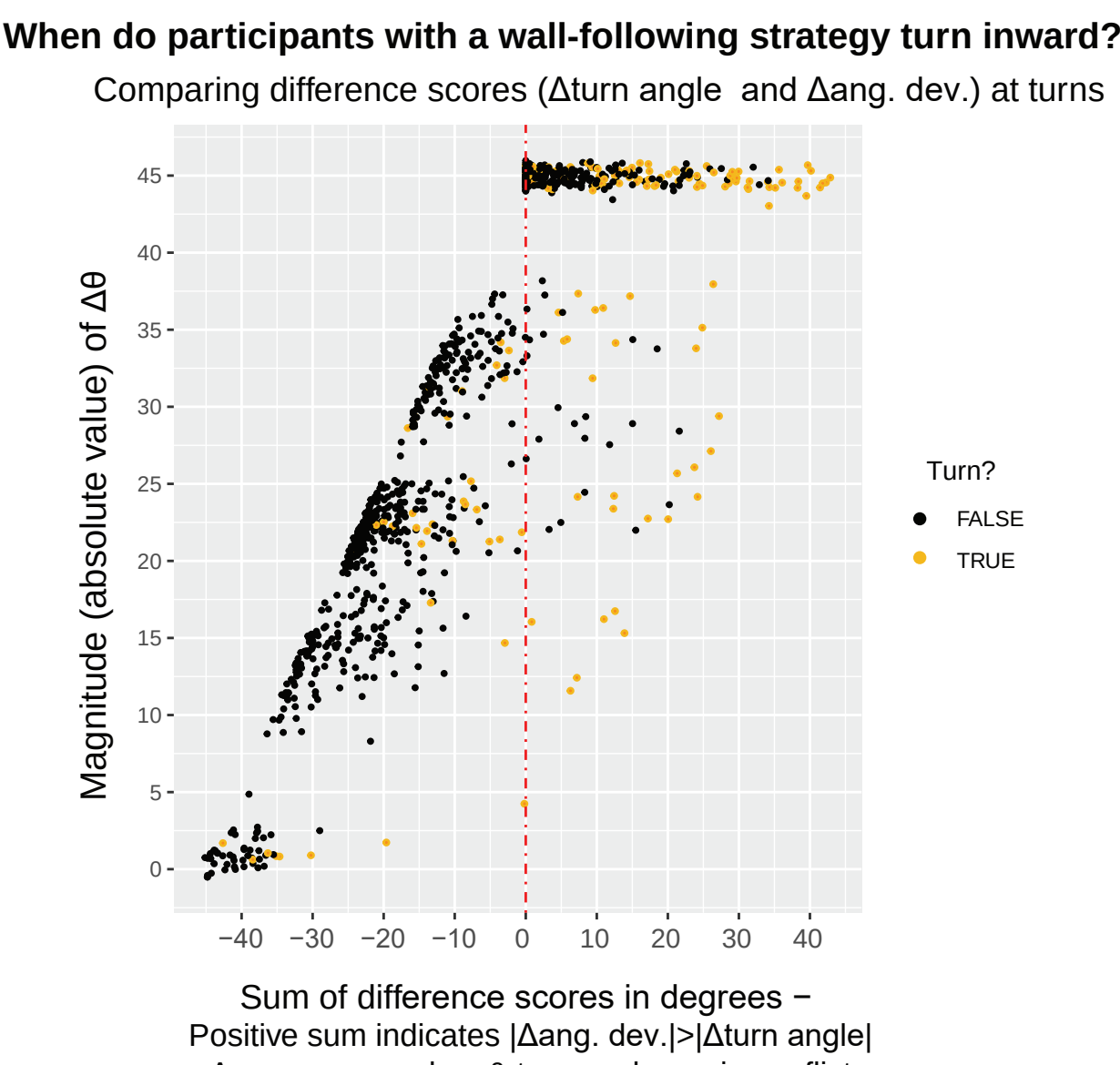
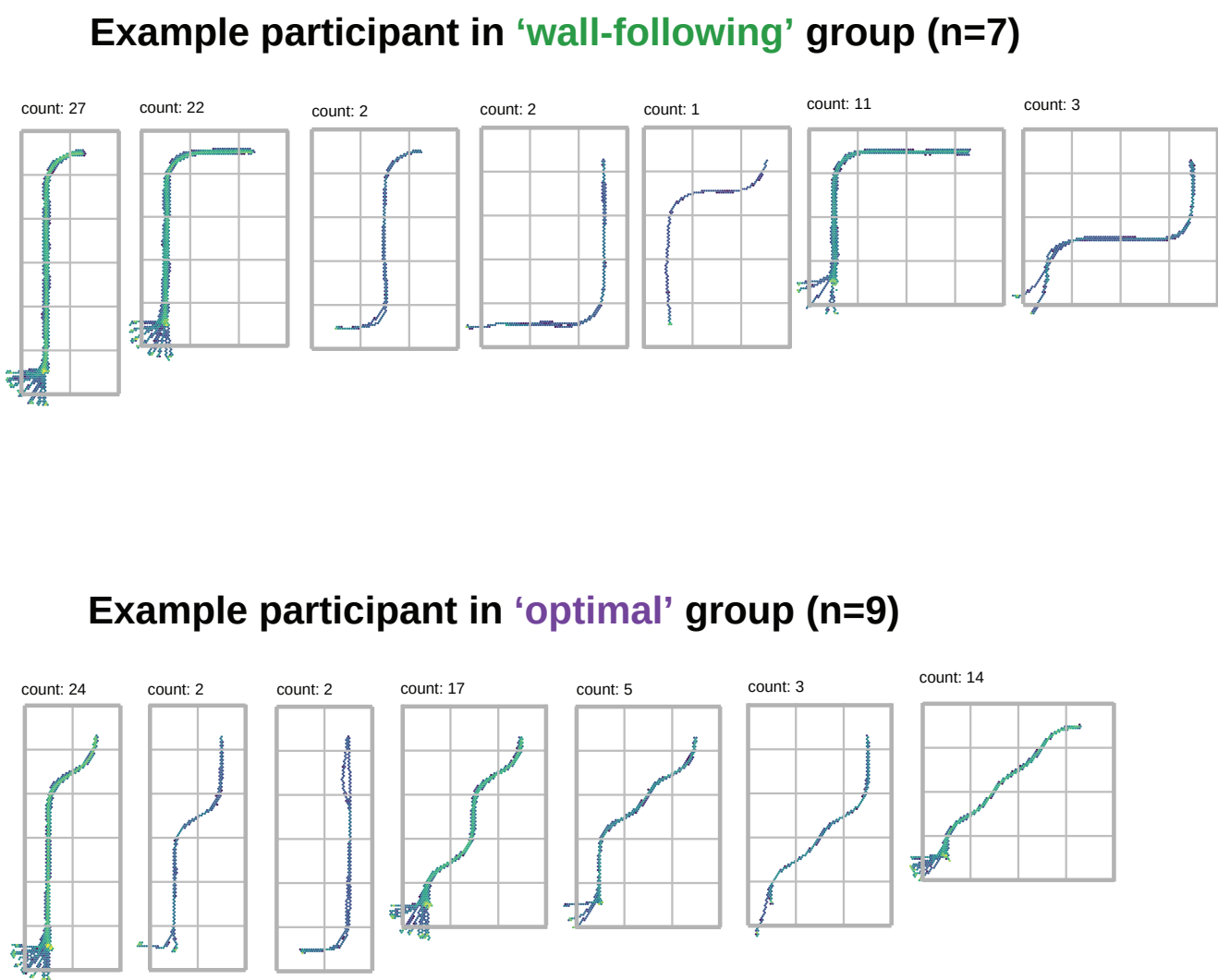
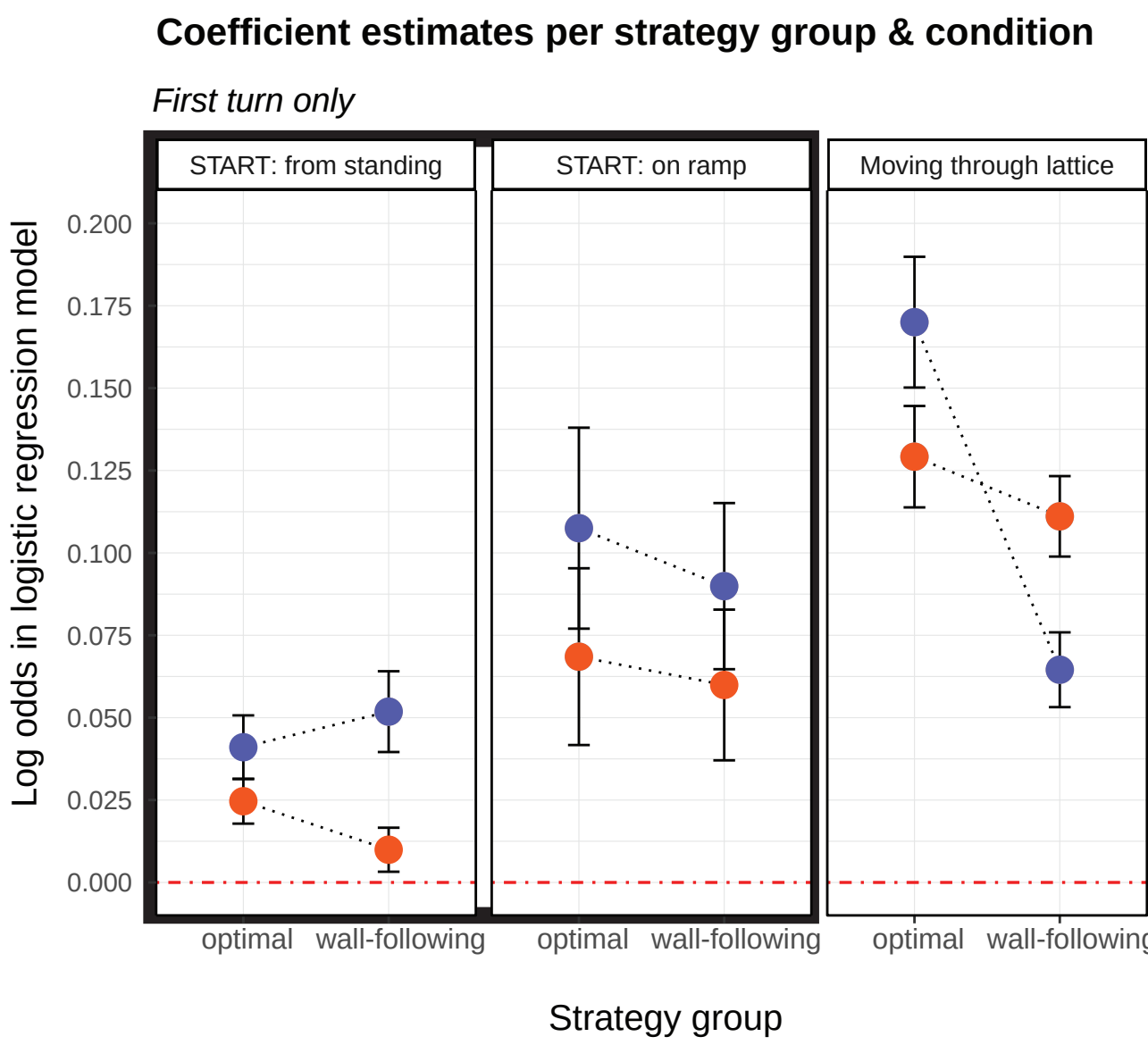
Results



The optimal route can be achieved by minimizing ang. dev. from beeline at doors



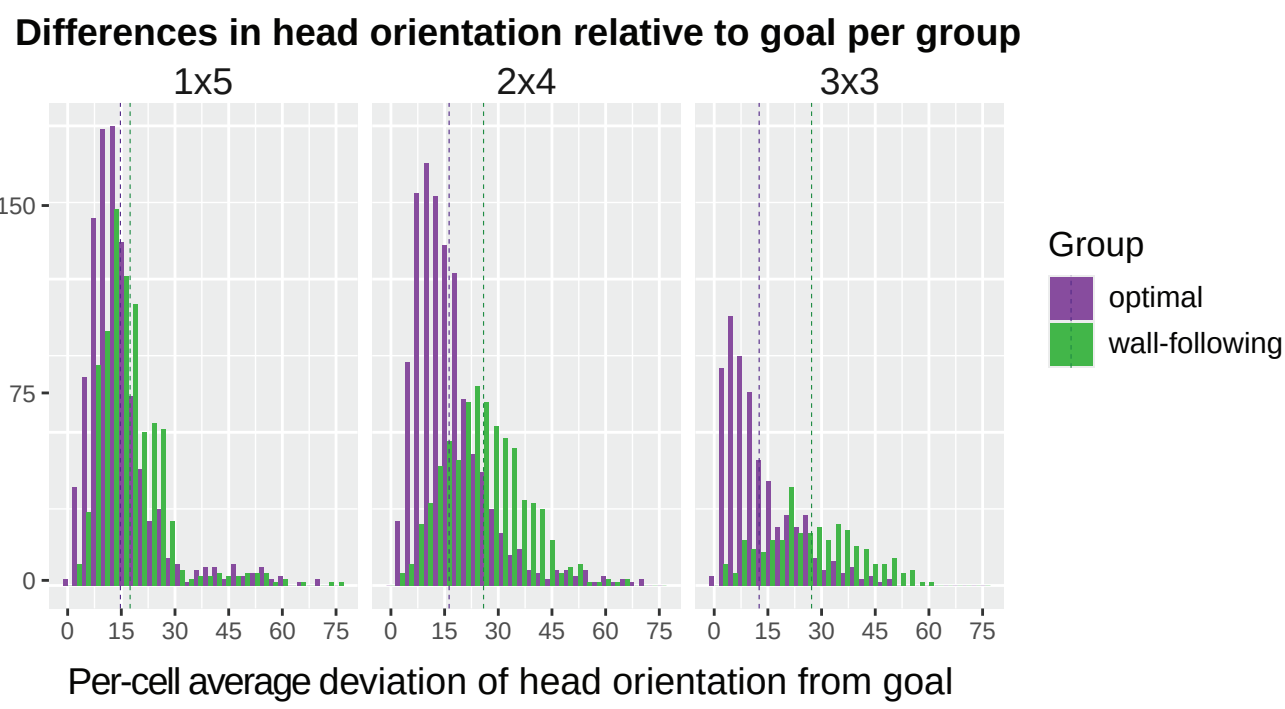
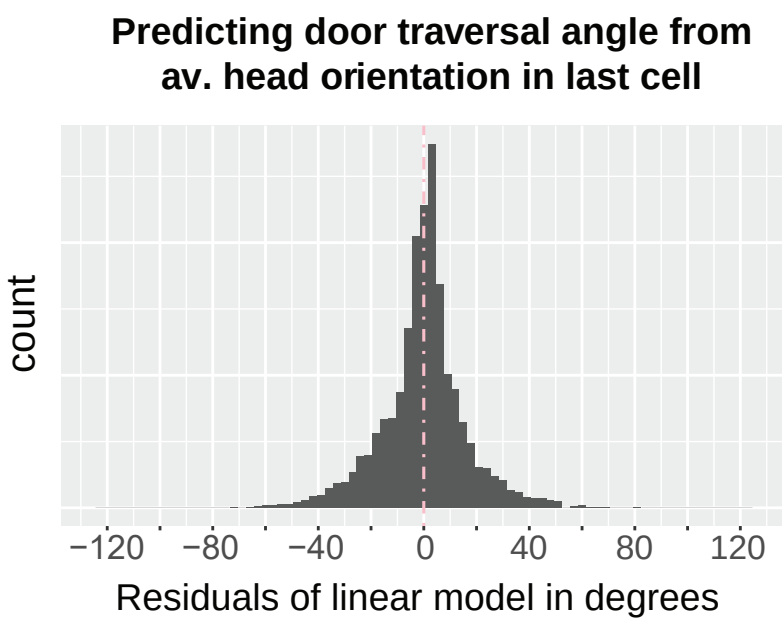
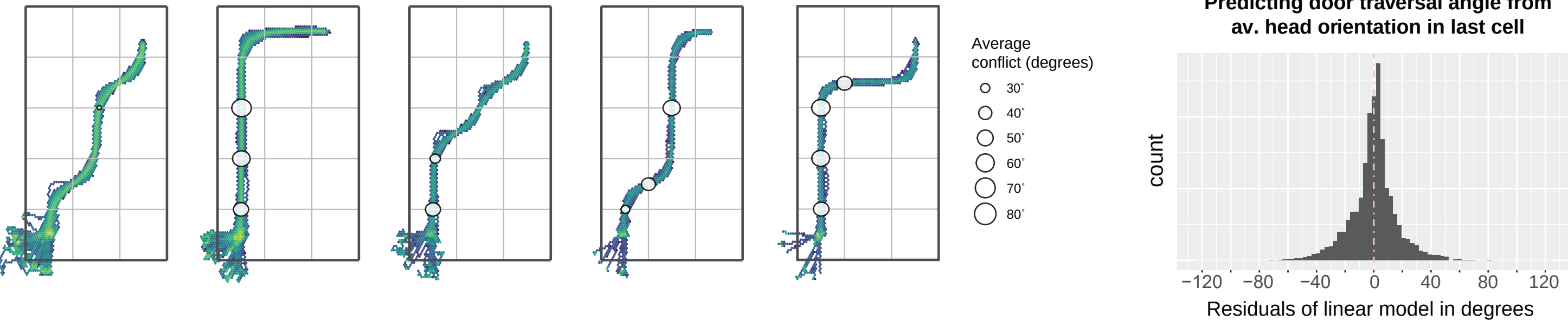
Routes are not uniformly likely. The most preferred are the near diagonal (energetically optimal) and the perimeter (cognitively simple)



The sample was divided based on whether participants used the optimal route >50% of the time

- Separately fitting log. reg. coefficients for deviation from beeline and turn angle by group predicts 92% of choices
- As expected, the wall-following group weighs turn angle more than deviation from beeline, while the optimal group has the opposite preference
- The first decision was consistent: in the first cell, both groups were more likely to move in the direction that minimized deviation from the beeline (long edge first)
- However, behavior was more random when participants started a trial from standing vs. entering the maze from an onramp
- The wall-following group occasionally turned into the maze interior; this was more likely when the angle of deviation from the beeline exceeded that of the turn

Quantifying conflicts between strategies (turn angle and beeline) at door thresholds for a sample of routes



- The optimal (near-diagonal) route reduces conflicts between ones deviation from the beeline and turn angle such that both can be minimized simultaneously
- This is achievable because the angle of traversal through doorframes is not perpendicular to the wall
- Angled travel through doors could arise from head orientation being biased towards the goal. In fact, head orientation in the last cell predicts door traversal angle

Discussion

Humans traversing rectangular grid-mazes either utilize a perceptual strategy that produces minimum-length, minimum-cost routes that approximate the diagonal or a strategy that involves following the perimeter and seldom turning into the interior, an approach with low cognitive cost. A model using only 2 variables, fit separately between these groups, explains 92% of choices. Starting conditions also affect route choice. All navigators prefer to start in the direction that fixes the goal closer to the center of their FOV, i.e.: the long edge. We did not find this in an earlier open courtyard experiment, suggesting occlusion of the path may promote this bias. Lastly, starting from standing within the maze, rather than entering while in motion, seems to make turn decisions more random.

References

[1] Engstrom, C., Srinivasan, M., Warren, W.H., (2025) Human navigators employ local strategies that approximate globally optimal routes under uncertainty (In Preparation).

[2] Baxter, B. A., & Warren, W. H. (2020). Route selection in barrier avoidance. *Gait & posture*, 80, 192–198.

[3] Brown, G. L., Seethapathi, N., & Srinivasan, M. (2021). A unified energy-optimality criterion predicts human navigation paths and speeds. *Proceedings of the National Academy of Sciences of the United States of America*, 118(29), e2020327118.